A Seeded Thirteen-Integer Jones Equation with Selective Sparse Carry Coding

Mathematical research artifact for ProveIt

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1 Thirteen integers: a seeded guard stack and a calibrated carry block

The polynomial specified below has exactly thirteen unrestricted integer unknowns, exact total degree 57 866 184, and a division-free circuit of 759 live binary operations (520 multiplications, 215 additions and 24 subtractions). Its integer zero existence is equivalent to a finite syntactic ZFC refutation. Thus, with the stated inherited compiler theorem, ZFC proves that its lack of integer zeros is equivalent to Con(ZFC). This is an ordinary mathematical proof, not a new Lean certificate, a proof of consistency, or a global minimality claim. Unlike the ten-integer companion, the arithmetic finalizer here uses only the established Jones ratio interface, not Sun's 2026 bridge.

1.1 A coherent fixed source and an explicit polynomial

Let $p_1, \dots, p_{22}$ be the original package-31 quadratic residuals after substituting $x_i = n_i + 1$ in all 31 positive source coordinates. The exact source is `upstream/parent_quartic31.json`; the complete fixed port mapping is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING.json`. All four compressed little-endian payloads are included and hash-authenticated. This uses the original transport coordinate, not package 45's extra transport shift. The imported source theorem says

$$\exists n \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} p_j(n) = 0 \quad \Longleftrightarrow \quad \text{there is a finite ZFC refutation.}$$

The fifth residual equals 1 at $n = 0$, so any common zero is nonzero.

Use the selective weights w_i, row order, and targets q_j in `CODE.json`, identical to the companion's certified tuple. Their maxima are $W = 183$ and $Q_* = 5190$; put $K_* = Q_* + 2W = 5556$ and $E_* = K_* + W + 2 = 5741$. For $p_j(n) = \sum_\alpha p_{j,\alpha} n^\alpha$, set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \qquad D_c(B) = \sum_{j,\alpha} c_{j,\alpha} B^{q_j - \alpha \cdot w}.$$

The finite isolation certificate checks all 71 808 possible interactions of the 136 used source terms with the 528 padded quadratic monomials. It proves

$$[t^{q_j}]\Big(1 + \sum_i n_i t^{w_i}\Big)^2 D_c(t) = 2p_j(n).$$

It also proves that the displayed exponents in D_c are distinct and its leading coefficient is 2. All $|c_{j,\alpha}| < A_* = 2^{2134927}$; the verifier derives this bound from the actual fixed-port bit lengths.

The unknowns are

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

Every subsequent symbol is a polynomial register, not an additional unknown. Fix

$$\kappa = 2134940, \quad a = 2^{\kappa} = 8192A_*, \quad \eta = 2^{16012040} = 2^{15\kappa/2-10},$$

and define

$$\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
N_0 &= B^{W+1}, & L &= 2bB^{E_*},\\
N &= \eta B^{2966}, & c &= bB^W,\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2(1+g)^2 D_c(B) + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c), & G_c &= 32c.
\end{aligned}$$

All powers have fixed nonnegative integer exponents. In particular, N is an ordinary polynomial, not a variable-exponent abbreviation.

Keep the literal Jones registers, with $\omega = w^2$:

$$\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R+1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= B_1 + jC, & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= JF, & t &= QF + J(H - C).
\end{aligned}$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s-1, 3\phi+2, 3\Delta-1).$$

The new seeded stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (1 \le j \le 3), \qquad C_* = C_3.$$

Finally define

$$\begin{aligned}
Z &= \epsilon w i z_1 z_2 z_3 a_0, & \Sigma &= Zu^5, & \rho &= t + ur,\\
\Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta,\\
A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 C_*^2 a_1.
\end{aligned}$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{1}$$

The literal division-free DAG is `generated/signed_seeded.json`.

1.2 Sparse padding and unconditional size estimates

Assume first only $b \geq 2$. Since $B - 1 > A_*$ and the top coefficient of D_c is 2, we have $D_c(B) > B^{Q_*} > 0$. Also $0 < M_c < N_0$. For every $g \geq 1$, it follows that $S_c, T_c > 0$ and

$$R = L^2 + S_c(L-1) + T_c L + L - 1 > L^2.$$

Moreover

$$N^2 = 2^{-20} a^{15} B^{5932} < B^{5947} < B^{11482} < L^2, \qquad b \leq c \leq N.$$

Thus $R \geq N \geq 8$ and $b \leq N, R$ before assuming any carry or square condition. Negative powers here occur only in this comparison proof, never in the equation.

If $1 \leq g < 4bB^W$, integrality gives $1 + g \leq 4bB^W$ and $g < N_0$. The injective coefficients imply

$$|D_c(B)| < \frac{A_* B^{Q_*+1}}{B-1}, \qquad 2(1+g)^2 D_c(B) < \frac{32 A_* b^2}{B-1} B^{K_*+1}.$$

Because $B = 8192 A_* b^2$, the last prefactor is less than $1/128$. Together with $T_* < B^{Q_*+1}/(B-1)$ this gives $0 < J_c < B^{K_*+1}$. Likewise $(B-2)T_* < B^{Q_*+1} \leq B^{K_*+1}$. Consequently

$$0 < S_c, T_c < B^{E_*}, \qquad S_c + T_c < 2B^{E_*} < L.$$

These estimates hold for arbitrary integer $b \geq 2$, not just powers of two.

Now suppose $b = 2^k$. Then $B = 2^\nu$, where $\nu = \kappa + 2k$. The binary mask M_c forces exactly

$$g \,\&\, M_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i}, \qquad 0 \leq n_i < b.$$

Here & is used only in the proof. For such a code, write $h_e = [t^e](1 + \sum n_i t^{w_i})^2 D_c(t)$. Then

$$|h_e| \leq A_*(1 + \textstyle\sum_i n_i)^2 < 1024 A_* b^2 = B/8 < B/4.$$

The unnormalized digits of J_c are $2h_e + \varepsilon_e$, where $\varepsilon_e = 1$ at a target and 0 elsewhere. Starting with incoming carry $c_0 = 0$, normalized carries obey

$$c_{e+1} = \left\lfloor \frac{2h_e + \varepsilon_e + c_e}{B} \right\rfloor, \qquad c_e \in \{-1, 0\}.$$

At a target, its normalized digit avoids $B - 2$ exactly when it is 0 or 1. If $h_e = 0$, the digit is $1 + c_e \in \{0, 1\}$. If $h_e > 0$, it is at least 2; if $h_e < 0$, a borrow normalizes it to a digit at least 2. Thus the target test is equivalent to $h_e = 0$, independently of incoming borrows. There is no need to fill every position with a dense $B/2$ offset. It follows, by splitting at N_0, that

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ codes a common zero of all } p_j.$$

Conversely every common zero has, for arbitrarily large $b = 2^{1+2r}$, such a code with $1 \leq g < bB^W = c$.

1.3 The parity-tuned two-block threshold

Write $L = 2^\lambda$ after $b = 2^k$ has been established. The three disjoint blocks in R give

$$s_2(R) = 1 + s_2(S_c + T_c) + \lambda - s_2(S_c) = \lambda + s_2(T_c) + 1 - \tau_2(S_c, T_c),$$

where s_2 is binary digit sum and $\tau_2(S, T) = s_2(S) + s_2(T) - s_2(S + T) \geq 0$ counts binary carries. The strict inequality $S_c + T_c < L$ is essential. The 31 allowed low positions and the 22 target positions give

$$s_2(T_c) = (W + 23)\nu - 31k - 22, \qquad \lambda = 1 + k + E_*\nu.$$

Therefore

$$\begin{aligned}\lambda + s_2(T_c) + 1 &= 5947\kappa + 11864k - 20 \\ &= 2\big(15\kappa/2 - 10 + 2966(\kappa + 2k)\big) = \log_2 N^2.\end{aligned}$$

Legendre's formula yields $v_2\binom{2R}{R} = s_2(R)$, hence

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

The small factor $2b$ in L calibrates the threshold to a polynomial square without an extra variable, an extra target mask, or a cubic radix.

1.4 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c-g) = c^2 - (c-2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\mathrm{aux}} = 8gK^2 + 1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\mathrm{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ and $F \equiv 1 \pmod{J}$ imply $F \mid H - C$. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put $d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $u \mid t$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U+1)^{2R}/U^R = Y + f$ with $0 \leq f < 2^{2R}/U = b/(2N^2)$ and $Y \geq 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \leq \frac{2(Y+f)}{Y(U+1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \geq 8$. Since $c, b \leq N$ and $g(c-g) \geq 1$, it follows that

$$\frac{c^2(C - KY)^2}{K^2} < \frac{c^2 b^2}{N^4} \leq 1 \leq g(c-g),$$

so the very same canonical witnesses satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

1.5 Seeded square recovery, with all zero branches excluded

For every integer assignment, $J = 4A - 5 \neq 0$ and $F \neq 0$. Indeed $F = 0$ forces $A = 0$ and $E = \pm 1$, whereas E is then divisible by $J = -5$. Thus $u = JF \neq 0$ unconditionally. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either its cubic has a simple root or that cubic is irreducible, whose Frobenius three-cycle makes its nonzero discriminant -23 a square; the quadratic then has a simple root. At 23 the cubic has the simple root 3. At 2, $-23 \equiv 1 \pmod 8$ has compatible square roots. Hensel lifting and CRT finish the assertion. Consequently $\Theta = u^5 \Pi(t/u + r) \neq 0$ for every integer assignment. Moreover

$$\exists r \quad Zu^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

For the reverse implication, choose a root of Π modulo $|Z|$ after $u \mid t$. For the forward implication, if a prime has $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contradicting $u^5 \mid \Theta$. If $Z = 0$, divisibility would demand $\Theta = 0$, already excluded.

At a zero of (1), $\Sigma = 0$ is impossible because then $\mathcal{F}_{13} = \Theta^4 \neq 0$. Therefore $\epsilon, w, i, z_1, z_2, z_3, a_0$ are all nonzero. Every $d_j \equiv 2 \pmod 3$ is nonzero, and every $B_j^* \equiv 1 \pmod 4$ is nonzero. Furthermore $B_j^* \neq 1, -1$, so $|B_j^*| \geq 3$ and $|C_*| > 1$ without any positivity hypothesis.

Let $q = X/\Sigma$. Equation (1) gives the exact identity

$$(q^2 + C_* - C_*^2 a_1)^2 = 4C_* q^2.$$

If $q = 0$, it would force $C_* a_1 = 1$, impossible since $|C_*| > 1$. Thus

$$r_0 = \frac{q^2 + C_* - C_*^2 a_1}{2q}, \qquad r_1 = \frac{q - r_0}{C_*}$$

are rational and satisfy $r_0^2 = C_*$, $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. This proof covers negative radicands and all possible cancellations without an implicit positivity or distinct-square-class assumption.

Now $\gcd(B_j^*, C_{j-1}) = 1$, so the nonzero factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime. Since their product C_* is a square, every absolute factor is a square. Each $B_j^* \equiv 1 \pmod 4$ is therefore positive, because a negative square cannot be 1 modulo 4. The sign of the product forces $a_0 > 0$, and a_0 is a square too. All prefixes are now positive. A negative d_j would make $B_j^* \leq -3$, so all $d_j > 0$, yielding $s \geq 1$, $\phi \geq 0$, and $\Delta > 0$. This argument does not assert that $a_0 \equiv 1 \pmod 4$; that false shortcut would invalidate the seeded construction.

Conversely, suppose a_0 is a positive square and these three sign conditions hold. Each d_j is positive and nonsquare since it is 2 modulo 3. Inductively C_{j-1} is a positive square, so $d_j C_{j-1}$ is positive nonsquare. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even; set $z_j = y/2 \neq 0$. Then $B_j^* = x^2$ and C_j is a positive square. Given also a_1 square and $u \mid t$, the intersective lemma chooses r with $\Sigma \mid \Theta$, and an integer choice of v makes $X/\Sigma = \sqrt{C_*} + C_* \sqrt{a_1}$. Thus (1) vanishes. Combining both directions proves the asserted equivalence to the original source and hence to a finite ZFC refutation.

1.6 Exact degree, circuit accounting, and provenance

For the seeded equation the relevant exact register degrees are

	R	N	a_0	a_1	C_*	Σ
deg	45932	11864	652210	4363426	5592732	6692101

and $\deg X = 6692102$, $\deg A_0 = 18976934$, $\deg A_1 = 28933092$. The last dominates X^2 and A_0, so the output's leading term is A_1^2. The symbolic diagonal certificate has leading coefficient

$$356960\, a^{14465376} \eta^{1888} \pmod{1000003},$$

which is nonzero because a and η are powers of two. The circuit upper bound and this surviving diagonal coefficient prove $\deg \mathcal{F}_{13} = 2(28933092) = 57866184$ exactly, not merely at most.

The metric counts every live binary addition, subtraction and multiplication, including scalar multiplications and every gate in fixed exponentiation. Fixed coefficient leaves and reuse are free; coefficient bit size is a separate measure.

The unseeded three-root baseline, also included, uses the same source, starts $C_0 = 1$, omits a_0 from Z, and combines the square conditions C_*, a_0, a_1 with the established separating weights $1, C_*, C_*(P^2 + K^2 + 2)$. It has 770 operations and degree 71382096. The seeded improvement saves eleven operations and reduces this degree by 18.93% without changing the thirteen unknowns. For completeness, if $W_2 = C_*(P^2 + K^2 + 2)$, its radicands are $A_0 = \Sigma^2 C_*$, $A_1 = \Sigma^2 C_*^2 a_0$ and $A_2 = \Sigma^2 W_2^2 a_1$. It uses

$$V = X^2 + A_2 - A_0 - A_1, \quad W' = V^2 + 4X^2 A_2 - 4A_0 A_1, \quad (W')^2 - 16V^2 X^2 A_2.$$

This is the product over all eight signs of $X - \Sigma(\pm\sqrt{C_*} \pm C_*\sqrt{a_0} \pm W_2\sqrt{a_1})$. The inequalities $|C_*| > \sqrt{|C_*|}$ and $\sqrt{|a_0|} \le P^2 + K^2 + 1$ make each weighted nonzero radical larger in complex absolute value than the sum of all preceding radicals. Thus effective sign patterns are distinct, also when a radicand is zero. A rational signed sum is fixed by every multiquadratic automorphism; uniqueness forces each nonzero root rational and hence integral. This recovers all three square conditions and the merger divisibility. At $\Sigma = 0$ its output is $\Theta^8 \neq 0$. The unseeded Pell stack and all remaining source and ratio steps are the same as proved above.

Run `python build_jones.py` and `python check_jones.py`. The checker authenticates the fixed payloads and pinned sources, independently replays the literal registers over three prime fields (including signed and zero inputs), verifies the formal norm identities, checks 136000 signed seeded stacks, and exercises nonzero merger and sparse-borrow branches. These finite tests corroborate the universal proofs, not replace them. All imported source paths, Git blob IDs, and SHA256 values are in `SOURCE_PINS.json`, at repository commit `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. The inherited compiler and Jones ratio theorems are explicitly identified; no newly compiled kernel certificate for this concrete equation is claimed.